BRACING CONNECTIONS

PURPOSE

This document provides standardized design criteria for bracing connections for which reference material may not be readily available. This document also provides a basis for further automation of the overall design process.

This document is also intended to promote the use of the *applicable* bracing details shown in Standards, 000.215.5050 & 000.215.5060, Structural Steel Standard Details - Bracing Connections.

SCOPE

This document covers the design of *NON-SEISMIC* bracing connections of the types shown in Figure 1 and Figure 2, with brace bolting configurations shown in Figure 3. Braces are bolted to gusset plates, and gusset plates are fully welded to supporting members on one edge. For combined connections, gusset plates and beams are bolted to a column using a single plate shear connection. Design criteria is included for both snug tight and slip critical bolting. This document includes a descriptive procedure and two sample designs.

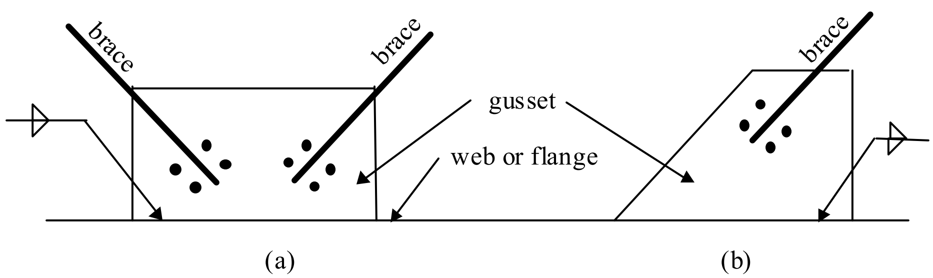

Figure 1: Simple Brace Connections

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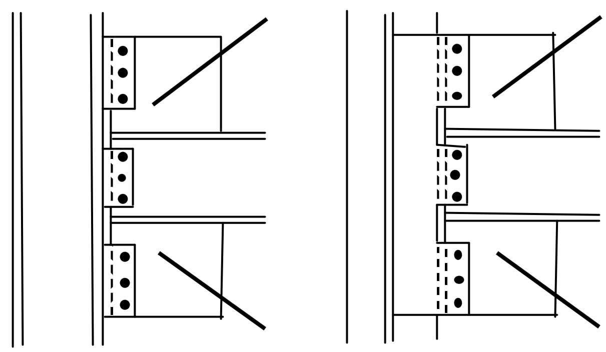
Figure 2: Combined Brace Connections



(a) (b)

Design of the brace, beam, or column is part of the overall structural analysis and is not covered by this document.

Details shown on the fabricator's shop drawings shall be checked to ensure they are consistent with design assumptions.

APPLICATION

This document applies to the design of braced steel structures. The design procedure uses the AISC Specification 13th Edition, LRFD method. AISC 13th edition is a major update with the combination of both Allowable Strength Design and Load and Resistance Factor Design methods. For Low-Seismic Applications, where design and detailing is not required in accordance with AISC 341, designs shall comply with AISC 360, as demonstrated by this guideline. For High-Seismic Applications, where design and detailing is required in accordance with AISC 341, designs shall comply with both AISC 341 and AISC 360 and is not covered in this guideline. For each of the limit states defined in this document, the following equation must be satisfied:

$$R_u \leq \phi R_n$$

Where,

R_u = factored load
ϕR_n = design strength

Load and Resistance Factor Design

If the structural analysis is performed based on Load and Resistance Factor design methods, loads at the connection must be factored to work with this practice. An effective way to factor loads is to obtain dead, live, and wind, and seismic forces from individual load cases and then manually apply the appropriate LRFD load factors. If individual load cases are not distinguishable, a single conservative load factor may be estimated and applied to the total load combination.

Design in accordance using Allowable Strength Design (ASD) is also permitted by AISC 360.

Bolting Configurations

There are several standard configurations for bolting braces to gusset plates as indicated in Figure 3. The following design procedure addresses these cases.

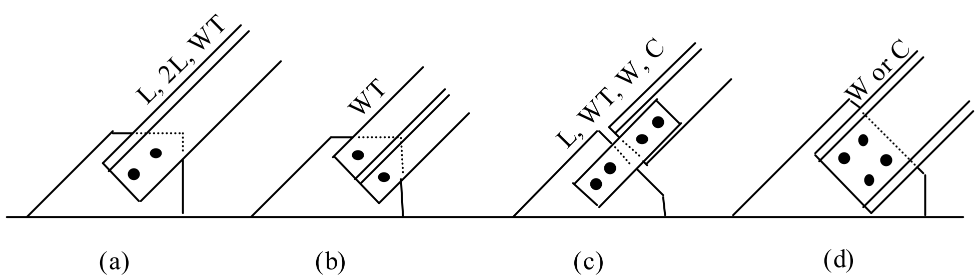


Figure 3: Bolting Configurations

DESIGN PROCEDURE – SIMPLE BRACE CONNECTIONS

The design of a simple bracing connection involves the following:

1. Determine which load cases are worthy of analysis. Note that some of the limit states listed below are specifically for brace tension or compression, or specifically for bearing-type or slip-critical connections.

2. Select trial component sizes and check the following limit states, each of which is described further in subsequent sections of this document.

   (a) Fracture on Net Section (tension loads)
   (b) Block Shear (tension loads)
   (c) Bolt Shear
   (d) Slip Resistance (if connection is slip-critical)
   (e) Bolt Bearing
   (f) Plate Buckling (compression loads)
   (g) Yielding on Gross Section of Plate (tension loads)
   (h) Weld Shear
   (i) Web Check (if connecting to a web)

4. Refine or revise trial component sizing for maximum effectiveness and efficient use of materials. Refer to the section "Refine/Revise Connection", later in this document.

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(a) Fracture on Net Section (tension loads)

This limit state involves a tensile fracture/rupture through a bolt row and is applicable to the brace, connecting plate, and gusset. The equations come from AISC Sections D2 and J4.1:

R_u = (factored tension force in brace) (kips) (Equation 1)

$\phi R_n = 0.75\ F_u\ A_e$ (kips) (Equation 2)

For connecting plates (Figure 3c), A_e is the area of both plates.

When a tension load is transmitted directly to each of the cross-sectional elements, the effective net area, A_e, is equal to the net area, A_n.

If the connection contains only one row of bolts perpendicular to the direction of loading, A_e is equal to the net area of the connected element (ex: if a WT section is bolted through its flange to a gusset plate with its web outstanding, only the net area of the flange shall be considered).

Otherwise, when a tension load is transmitted through some but not all of the cross-sectional elements, the effective net area is,

$A_e = A_n\ U$ (Equation 3)

$U = 1 - (x / L) \leq 0.9$ (Equation 4)

The connection eccentricity, x, is the distance from the centroid of the brace to the face of the gusset (refer to Table D3.1). For the double angle connection of Figure 3a, the connections eccentricity is the dimension from the gusset face to the centroid of either angle.

For a connection with only one bolt row, such as the WT connection of Figure 3b, use an effective net area, A_e, equal to the net area of the connected element. For the WT of Figure 3b, this would be the net area of the flange.

The AISC commentary provides values of the reduction coefficient, U, which may be used in lieu of Equation 4 (refer to Table D3.1).

In the preceding equations,

F_u = specified minimum tensile strength of brace, ksi
A_e = effective net area, in^2
A_n = net area of member, in^2 (refer to Section D3.2)
U = reduction coefficient (refer to Table D3.1)
x = connection eccentricity, in
L = length of connection in the direction of loading, in

If the design strength is not adequate, try a larger or thicker brace, or perhaps a different bolt arrangement to increase the effective net area.

(b) Block Shear (tension loads)

This limit state involves a combination of tensile fracture transverse to the direction of load and a shear failure parallel to the direction of load. This limit state applies to the brace, connecting plates, and gusset. Refer to Figure 4 for several graphical descriptions. Background information is provided in the AISC Commentary, Section J4. Due to variations in actual gusset plate fabrication preferences, minimum AISC specified bolt spacing and edge distances should be assumed unless specifically detailed otherwise.

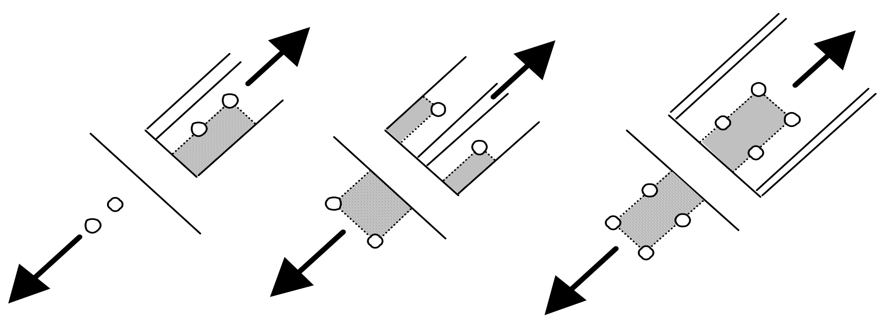

Figure 4: Block Shear Mechanisms

R_u = (factored tension force in brace) (kips) (Equation 5)

Resistance to block shear is based on the following two equations from the AISC Specification, Section J4. The equations represent shear rupture with tension yielding, and tension rupture and shear yielding.

ϕR_n= (Equation 7) $\leq$ (Equation 6)

$0.75\,[0.6\,F_y\,A_{gv} + U_{bs}\,F_u\,A_{nt}]$ (kips) (Equation 6)

$0.75\,[0.6\,F_u\,A_{nv} + U_{bs}\,F_u\,A_{nt}]$ (kips) (Equation 7)

Where,

F_u = specified minimum tensile strength of connected part, ksi
F_y = specified minimum yield strength of connected part, ksi
A_{gv} = gross area subject to shear, in^2
A_{nv} = net area subject to shear, in^2
A_{gt} = gross area subject to tension, in^2
A_{nt} = net area subject to tension, in^2

Where the tension stress is uniform, U_{bs}=1; where the tension stress is non-uniform,

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U_{bs}=0.5. See AISC 360 commentary, p. 16.1-351 for further explanation

For connecting plates, gross and net areas are for both plates.

If the design strength is not adequate, try increasing the plate thickness, or using a different bolt arrangement to alter the section subject to rupture.

(c) Bolt Shear

This limit state involves a shear failure of the bolt material and applies to bearing-type and slip-critical connections. The following is based on threading in the shear plane and a fastener pattern no greater than 50 inches in length. The design capacity is from AISC Table J3.2 and section J3.6.

R_u = (factored brace force) / N (kips / bolt) (Equation 8)

$\phi R_n = 0.75\ F_n\ (A_b)\ (N_s)$ (kips / bolt) (Equation 9)

Where,

A_b = nominal body area of bolt, in^2
F_u = specified minimum tensile strength of the bolt material, or shear stress, F_{nv} from Table J3.2, ksi
N = number of bolts
N_s = number of shear planes

If the design strength is not adequate, try additional bolts or larger diameter bolts.

(d) Slip Resistance (if connection is slip-critical)

This limit state involves a slippage between the surfaces of two plates clamped together by fully tensioned bolts. Bolts in slip-critical connections to be designed to prevent either a serviceability limit state or the required limit state, as stated in AISC 360, section J3.8. Equations for factored load option are presented here.

R_u = (factored brace force) / N (kips / bolt) (Equation 10)

$\phi R_n = \phi\ \mu\ D_u\ h_{sc}\ T_b\ N_s$ (kips / bolt) (Equation 11)

Where,

N = number of bolts
N_s = number of slip planes
T_b = minimum fastener tension, kips
μ = mean slip coefficient
D_u =1.13; a multiplier that reflects the ratio of the mean installed bolt pretension to the specified minimum bolt pretension.

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h_{sc} = hole factor

The mean slip coefficient, μ, is 0.35 for hot dip galvanized and roughened surfaces. For other surfaces, refer to Section J3.8.

The minimum fastener tension, T_b, is 39 kips for 7/8 inch diameter A325 bolts. For other sizes and grades, refer to Table J3.1.

The resistance factor, ϕ, refer to section J3.8 for applicable design slip requirements.

If the design strength is not adequate, try additional bolts or larger diameter bolts.

(e) Bolt Bearing

This limit state involves the bearing strength of the bolt hole and applies to bearing-type and slip-critical connections. If material strengths and hole sizes of the connected parts are the same, the thinner plate will control. The following criteria, a simplification of Section J3.10, apply to the most common case of standard holes and service load design consideration. For other hole types, refer to Section J3.10. Minimum hole spacing criteria is provided in Section J3.3 and minimum edge distance criteria in Section J3.4.

R_u = (factored brace force) / N (kips / bolt) (Equation 12)

Design strength is the lessor of the following two equations,

$\phi R_n = 0.75\ (1.2)\ (L_c)\ (t)\ (F_u)$ (kips / bolt) (Equation 13)

$\phi R_n = 0.75\ (2.4)\ (d)\ (t)\ (F_u)$ (kips / bolt) (Equation 14)

Where,

d = nominal bolt diameter, in
L_c = clear distance, in (refer to following note)
t = thickness of connected part, in
F_u = specified minimum tensile strength of connected part, ksi
N = number of bolts

The clear distance, L_c, is the dimension, in the direction of force, between the edge of the hole and the edge of the adjacent hole, or to the edge of the material. Individual bolts in a connection will have individual resistances. If need be, factored load and resistance may be determined based on a summation of the individual bolts.

For double angle connections (Figure 3a), the thickness, t, for the brace should be the sum of thicknesses for both angles. For connecting plates, the thickness, t, should be the thickness of both plates.

If the design strength is not adequate, try increasing the thickness, increasing the edge distance or hole spacing, using additional bolts, or a larger bolt diameter.

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(f) Plate Buckling (compression loads)

This limit state involves compression buckling of the connecting plate or gusset. The effective width in compression is based on a Whitmore section. Unless limited by plate dimensions, the width of a Whitmore section is computed from the bolt spacing with a spread angle of 30 degrees from the first bolt row to the last. Refer to Figure 5 for a graphical description. The following criteria are from Section E3 and 9-3 to 9-4.

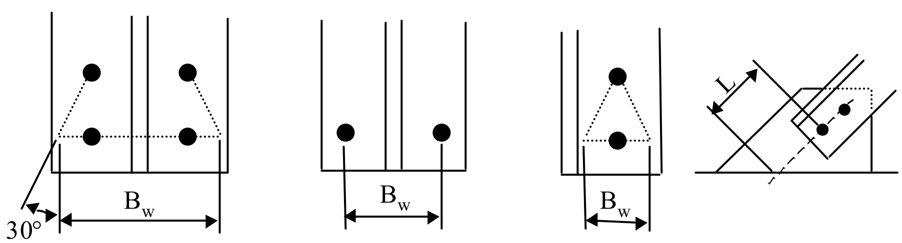

Figure 5: Whitmore Section Width, B_w, and Length, L

R_u = (factored compression force in brace) (kips) (Equation 15)

$\phi R_n = 0.90\,(A_g)(F_{cr})$ (kips) (Equation 16)

Where,

A_g = gross area of Whitmore section, in^2
F_{cr} = critical buckling stress from Section E3, ksi

For connecting plates, A_g is the area of both plates.

If part of the Whitmore section falls within the supporting member's web, then the gross area, A_g, should be adjusted accordingly. Refer to the example in AISC, page 9-4.

In determining F_{cr}, KL/r may be determined based on a K value of 1.2. This K value is based on a gusset supported on one edge. The buckling length, L, is the distance, measured in the direction of force, from centerline of last bolt to the surface of the supporting member (refer to Figure 5).

If the design strength is not adequate, try a thicker gusset plate, or a larger spacing between bolts to widen the effective plate width.

(g) Yielding on Gross Section of Plate (tension loads)

This limit state involves yielding of the connecting plate or gusset. Refer to the preceding AISC section for details, Section D2.

R_u = (factored tension force in brace) (kips) (Equation 17)

$\phi R_n = 0.9\,(F_y)(A_g)$ (kips) (Equation 18)

Where,

A_g = gross area, in^2
F_y = specified minimum yield stress of gusset, ksi

For connecting plates, A_g of a member is the total cross-sectional area.

If the design strength is not adequate, try a thicker gusset plate, or a larger spacing between bolts to widen the effective plate width.

(h) Weld Shear

This limit state involves a fillet weld shear failure. Refer to Figure 6 for a graphical description of the dimensions involved. The following criteria, from Section J2.4, is based on a fillet weld on each side of the gusset plate. Minimum weld size criteria is provided in Table J2.4.

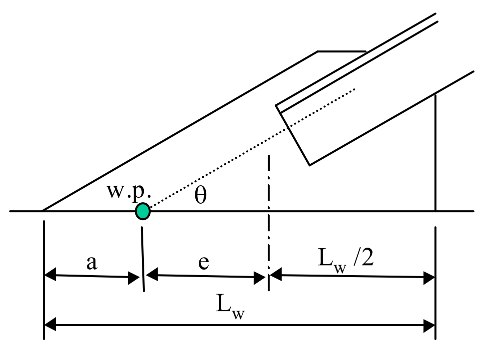

Figure 6: Moment Determination on Weld

$$f_a = P_u\,(\sin\theta) / 2\,(L_w) + P_u\,(\sin\theta)(e)\,3 / (L_w)^2 \qquad \text{(Equation 19)}$$

$$f_v = P_u\,(\cos\theta) / 2\,(L_w) \qquad \text{(Equation 20)}$$

$$R_u = \sqrt{(f_v)^2 + (f_a)^2} \quad \text{(k/in)} \qquad \text{(Equation 21)}$$

$$\phi R_n = 0.75\,(0.6\,F_{exx})(0.707\,t_w) \quad \text{(k/in)} \qquad \text{(Equation 22)}$$

Where,

P_u = factored load in brace, kips
F_{exx} = classification strength of weld metal, ksi
t_w = size of fillet weld, in
f_v = shear component of weld load, k/in
f_a = axial component of weld load, k/in
θ = angle of brace from horizontal, degrees

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L_w = length of weld, in
e = eccentricity of load at center of weld length, in

If the design strength is not adequate, try a larger weld size, or a longer weld length.

(i) Web check

This limit state applies when connecting into a beam or column web and involves a bending failure in the web due to the out-of-plane force from the gusset. The following criteria, from Kapp, assumes the sides of the web are restrained by the flanges:

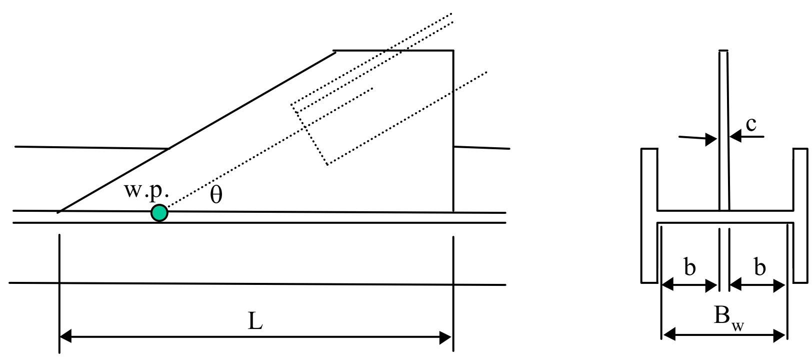


Figure 7: Web Criteria Dimensions

$b = (B_w - c)/2$ (in) (Equation 23)

R_u = (factored brace force)(sin θ) (kips) (Equation 24)

$\phi R_n = 0.9(F_y)(t_w)^2\left[L/b + 4\sqrt{1+(c/2b)}\right]$ (kips) (Equation 25)

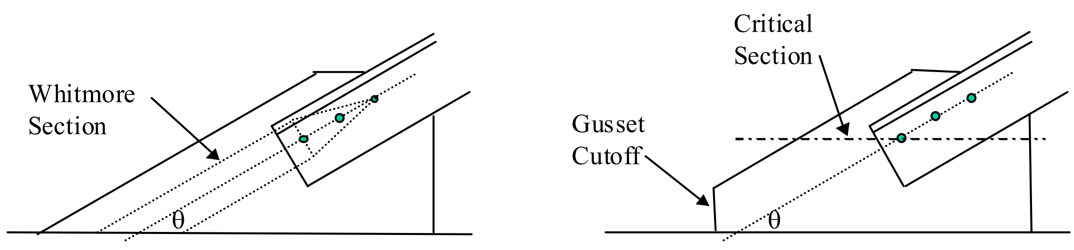
Where,

B_w = web width between flanges, in
θ = angle of brace from horizontal, degrees
F_y = yield strength of web, ksi
t_w = web thickness, in
L = length of plate, in
b = yield line dimension, in
c = thickness of gusset, in

The effect of another brace connection on the opposite side of the web should be included.

If the design strength is not adequate, increase the length of the gusset, or add stiffener plates to the column.

Plate Shear



(a) (b)

This limit state involves a shearing failure of the gusset plate. Normally this limit state is not a problem because the Whitmore section, by itself, is capable of transmitting the brace load axially without any shear (Figure 8a). In reality, some shear is generated due to the distribution of brace load into the actual dimension of the gusset plate. The shear load might become significant if the end of the gusset plate is cut off (Figure 8b) due to a low brace angle. The critical section is at the location of minimal gusset plate width, usually at the first brace bolt as indicated in Figure 8b).

Figure 8: Gusset Shear

R_u = (factored brace load)(cos θ) (kips) (Equation 26)

$\phi R_n = (0.9)(0.6)(F_y)(A_g)$ (kips) (Equation 27)

Where,

θ = angle of brace from horizontal, degrees
F_y = yield strength of gusset plate, ksi
A_g = area of gusset plate at the first brace bolt, in^2

If the design strength is not adequate, try a thicker gusset plate.

(h) Gusset Shear

This limit state involves a shear failure of the gusset plate near the beam weld or near the single plate column connection. Because the weld will be applied to the entire length of the gusset plate, shear rupture at the weld need not be checked. The following criteria is from Section J4.2:

R_u = factored shear, H_b or V_c (kips) (Equation 37)

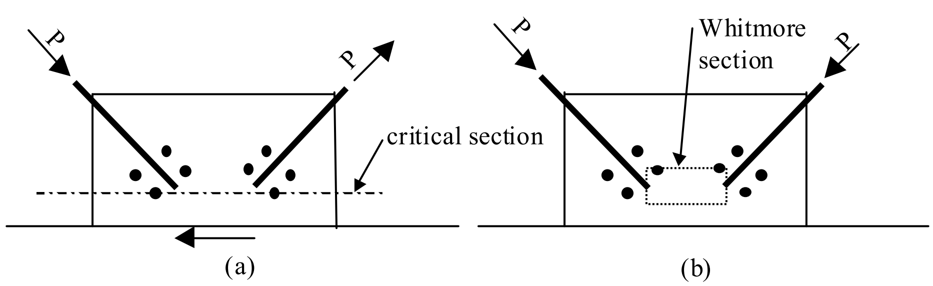
$\phi R_n = 0.9\ (0.6\ F_y)\ (A_g)$ (kips) (Equation 38)

Where,

F_y = yield strength of gusset material, ksi
A_g = area of gusset resisting shear, in^2

If the design strength is not adequate, try increasing the thickness of the gusset, or lengthening the gusset plate.

Two Braces on a Gusset



(Figure 1a). The following represents a logical extension to the single brace procedure.

Figure 9: Double Brace Connections

- The braces and gusset should first be checked as though the gusset plates were separate.
- For opposing brace forces (Figure 9a), gusset shear should be checked across the width of the combined gusset plate.
- For matching brace forces (Figure 9b), gusset buckling and/or yielding should be checked on a Whitmore section assumed to extend between the two braces.

DESIGN PROCEDURE – COMBINED BRACE CONNECTIONS

The design of a combined brace connection involves the following steps:

1. Determine which load cases are worth of analysis. Note that some of the limit states listed below are specifically for brace tension or compression, or specifically for bearing-type or slip-critical connections.

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2. Determine the distribution of forces between beam and column. Refer to the following section on the Uniform Force Method.

3. Select trial component sizes and check the following limit states for the brace-gusset-beam connection. Gusset shear is described in a following section. The remaining limit states are described within Design Procedure - Simple Brace Connections.

   (a) Fracture on Net Section of Brace (tension loads)
   (b) Block Shear on Brace or Gusset (tension loads)
   (c) Bolt Shear
   (d) Slip Resistance (if connection is slip-critical)
   (e) Bolt Bearing on Brace or Gusset
   (f) Gusset Buckling (compression loads)
   (g) Gusset Yielding on Gross Section (tension loads)
   (h) Gusset Shear
   (i) Gusset-Beam Weld Shear

4. Refer to Guideline 000.215.1201, Single Plate Connections to check each of the column shear plates for the loads determined in step 2.

   - Connecting into a column flange (Figure 2a) is straightforward. The shear plates are separate.

   - Connecting into the web (Figure 2b) is a bit more complex in that the plates are connected. The design can be handled by treating the plates as separate extended shear plates.

   If a shear plate design cannot be made to work, then column stiffener plates can be added in order to design the shear plate and stiffeners as a short wide flange cantilever beam.

5. Refine or revise trial component sizing for maximum effectiveness and efficient use of materials. Refer to the section "Refine/Revise Connection", later in this document.

Uniform Force Method

Determine the distribution of brace forces between the beam connection and the column connection. Background is provided in AISC page 13-3.

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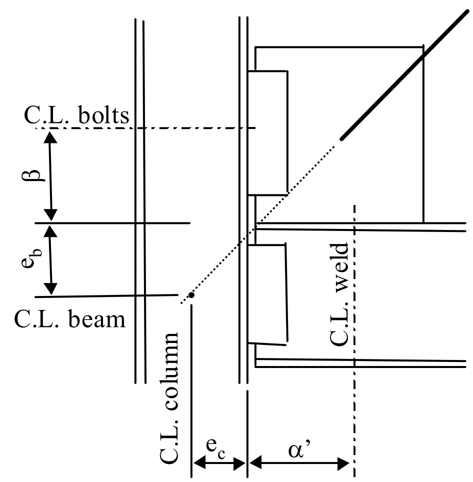

Figure 10: Uniform Force Method Dimensions

Determine the following required parameters from connection geometry:

- β = distance from center of column connection to top of beam, in
- α' = distance from center of beam connection to face of column, in
- θ = angle of brace from horizontal, degrees
- e_c = horizontal distance from work point to face of column, in
- e_b = vertical distance from work point to top of beam, in

Because the beam connection is considered more rigid,

$$\alpha = e_b \tan\theta - e_c + \beta \tan\theta \qquad \text{(Equation 28)}$$

$$r = \sqrt{(\alpha + e_c)^2 + (\beta + e_b)^2} \qquad \text{(Equation 29)}$$

The forces distributed to the beam and the column are:

$$H_b = \alpha\, P_u / r \qquad \text{(Equation 30)}$$

$$V_b = e_b\, P_u / r \qquad \text{(Equation 31)}$$

$$M_b = V_b\, (\alpha - \alpha') \qquad \text{(Equation 32)}$$

$$H_c = e_c\, P_u / r \qquad \text{(Equation 33)}$$

$$V_c = \beta\, P_u / r \qquad \text{(Equation 34)}$$

Where,

α = modified distance from beam connection to column, in

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P_u = factored load in brace, kips (tension is positive)
r = diagonal distance from w. p. to intersection of connection centers, in
H_b = factored horizontal force applied to gusset-beam connection, kips
V_b = factored vertical force applied to gusset-beam connection, kips
M_b = factored moment applied to gusset-beam connection, kips
H_c = factored horizontal force applied to gusset-column connection, kips
V_c = factored vertical force applied to gusset-column connection, kips

The design forces at the interface of beam and column become:

$$V_w = V_g - V_b \quad \text{(Equation 35)}$$

$$H_w = H_g + H_b \quad \text{(Equation 36)}$$

Where,

V_g = factored beam shear, kips (a downward at the support is positive)
H_g = factored beam axial force, kips (tension is positive)
V_w = factored vertical (shear) force applied to beam-column connection, kips
H_w = factored horizontal (axial) force applied to beam-column connection, kips

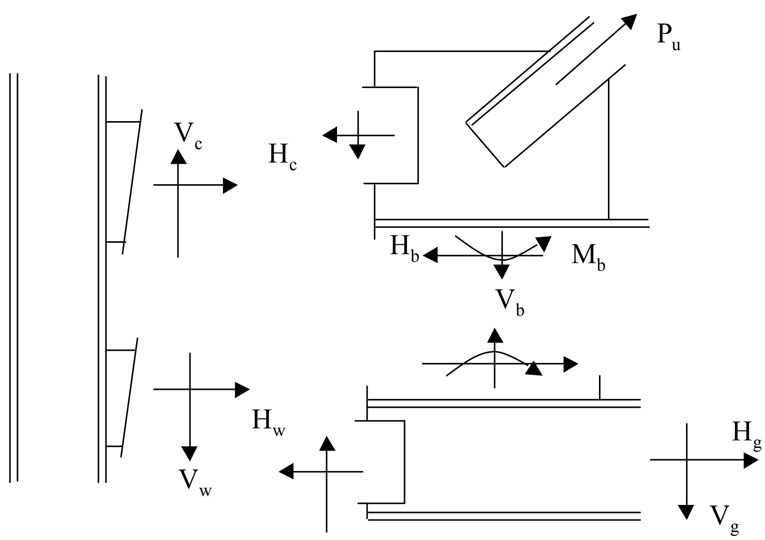

Figure 11: Positive Direction of Forces

REFINE/REVISE CONNECTION

The overall design should be reviewed for constructability and cost effectiveness.

- Review unity checks for all limit states. Ideally, more than one limit state should be very close to limiting values. Otherwise, it may be desirable to make adjustments to

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the design.

- Use no more bolts than calculations indicate as necessary. Additional bolts will only serve to increase field labor.

- Confirm that the gusset plate size chosen minimizes the length and size of the weld.

REFERENCES

Steel Construction Manual, Thirteenth Edition, American Institute of Steel Construction, Inc., 2005

Seismic Design Manual, Second Edition, American Institute of Steel Construction, Inc. and Structural Steel Educational Council, 2006

Guide to Design Criteria for Bolted and Riveted Joints, Second Edition, Geoffrey L. Kulak, John W. Fisher, and John H. A. Struik, John Wiley & Sons, New York, 1987

Steel Structures: Design and Behavior, Charles G. Salmon and John E. Johnson, Third Edition, Harper Collins, New York, 1990

Yield Line Analysis of a Web Connection in Direct Tension, Richard A. Kapp, *AISC Engineering Journal*, American Institute of Steel Construction, Second Quarter, 1974, pp 38-40

Handbook of Structural Steel Connection Design and Details, Akbar Tamboli (editor), Mc Graw Hill, New York, 1999

Structural Engineering
Guideline 000.215.1201: Single Plate Connections

Structural Engineering
Practice 000.215.5050: Structural Steel Standard Details - Bracing Connections

ATTACHMENTS

Attachment 01:
Sample Design – Simple Brace Connection

Attachment 02:
Sample Design – Combined Brace Connection

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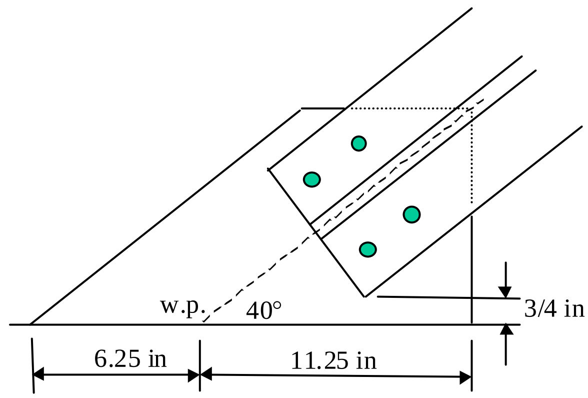
Sample Design - Simple Brace Connection

Given:



Brace Load:
to illustrate each section of the procedure,
let P_u = 80.0 kips tension or 60 kips compression

Brace: WT6x20, A572, Grade 50
$A = 5.84\ in^2$
$b_f = 8.01$ in
$t_f = 0.515$ in
$y = 1.09$ in
$F_y = 50$ ksi
$F_u = 65$ ksi

Note that the clearance dimension of ¾" could be larger, if necessary, as the gusset is connected into the web of a wide flange section.

Support: The gusset connects into the web of a W8x31, A572, Grade 50
$d = 8$ in
$t_w = 0.285$ in
$t_f = 0.435$ in

Try:

4 - 7/8 in diameter, A325 bolts on gage = 5.0 in (based on minimum edge distance)
bearing-type connection, threads in shear plane, standard size holes
1.5 in minimum from centerline of bolt to edge of plate (AISC minimum for a sheared edge is 1 ½ in)
3 in bolt spacing (AISC minimum is 2 1/3 in)
$A_b = 0.601\ in^2$
$F_u = 120$ ksi

Gusset: 3/8 in thick, A36
$F_y = 36$ ksi
$F_u = 58$ ksi

¼ in fillet weld on each side of plate
E70XX electrodes, $F_{exx} = 70$ ksi

Fracture on Net Section of Brace

factored tensile load, (Equation 1)
$R_u = P_u = 80.0$ kips

connection eccentricity,

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

x = (WT axis dimension, y) = 1.08 in

reduction factor,
$U = 1 – (x / L) = 1 – (1.08 \text{ in}) / (1 \text{ space})(3 \text{ in, bolt spacing}) = 0.64$

hole size + 1/16, (LRFD Section B2)
d_h = bolt diameter + 1/16 inch + 1/16 in= (7/8 in) + 2 (1/16 in) = 1.0 in

net area, (LRFD code, Section B3)
$A_n = (A) – (2 \text{ holes})(d_h)(t_f) = (5.89 \text{ in}^2) – 2 (1.0 \text{ in})(0.515 \text{ in}) = 4.86 \text{ in}^2$

effective area, (Equation 3)
$A_e = U (A_n) = (0.64)(4.86 \text{ in}^2) = 3.11 \text{ in}^2$

tensile resistance, (Equation 2)
$\phi R_n = 0.75\, F_u\, A_e = (0.75)(65 \text{ ksi})(3.11 \text{ in}^2) = 151.6 \text{ kips}$

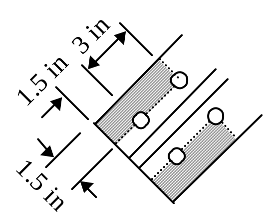
Unity Check = $(R_u) / (\phi R_n) = (80.0 \text{ kips}) / (151.6 \text{ kips}) = 0.53 < 1.0$, OK

Block Shear on Brace

factored tensile load, (Equation 5)
$R_u = P_u = 80.0 \text{ kips}$

gross shear area,
A_{gv} = [(3.0 in bolt spacing) + (1.5 in edge distance)] (2 sides)(0.515 in flange) = 4.64 in^2

net shear area,
A_{nv} = 4.64 in^2 – (1.5 holes)(1.0 in diameter)(2 sides)(0.515 in flange) = 3.10 in^2



gross tension area,
A_{gt} = [(8.005 in flange width) – (5.0 in gage)](0.515 in flange) = 1.55 in^2

net tension area,
A_{nt} = 1.55 in^2 – (½ hole)(1.0 in diameter)(2 sides)(0.515 in flange) = 1.04 in^2

shear yielding, tension rupture, (Equation 6)
$\phi R_n = 0.75\,[0.6\, F_y\, A_{gv} + F_u\, A_{nt}] = 0.75\,[(0.6)(50 \text{ ksi})(4.64 \text{ in}^2) + (65 \text{ ksi})(1.04 \text{ in}^2] = 155.1 \text{ kips}$

shear rupture, tension yielding, (Equation 7)
$\phi R_n = 0.75\,[0.6\, F_u\, A_{nv} + F_u\, A_{nt}] = 0.75\,[(0.6)(65 \text{ ksi})(3.10 \text{ in}^2) + (65 \text{ ksi})(1.04 \text{ in}^2)] = 141.4 \text{ kips}$

using the lower limit,
$\phi R_n = 141.4 \text{ kips}$

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Sample Design - Simple Brace Connection

Unity Check = $(R_u) / (\phi R_n)$ = (80.0 kips) / (141.4 kips) = 0.57 < 1.0, OK

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

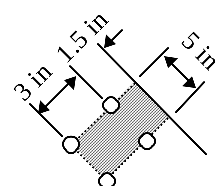
Block Shear on Gusset

factored tensile load, (Equation 5)
$R_u = P_u = 80.0$ kips

gross shear area,
$A_{gv} = [(3.0 \text{ in spacing}) + (1.5 \text{ in edge distance})] (2 \text{ sides})(0.375 \text{ in plate}) = 3.38 \text{ in}^2$

net shear area,
$A_{nv} = 3.38 \text{ in}^2 – (1.5 \text{ holes})(1.0 \text{ in diameter})(2 \text{ sides})(0.375 \text{ in flange}) = 2.26 \text{ in}^2$

gross tension area,
$A_{gt} = (5.0 \text{ in gage})(0.375 \text{ in flange}) = 1.88 \text{ in}^2$



net tension area,
$A_{nt} = 1.88 \text{ in}^2 – (½ \text{ hole})(1.0 \text{ in diameter})(2 \text{ sides})(0.375 \text{ in flange}) = 1.51 \text{ in}^2$

shear yielding, tension rupture, (Equation 6)
$\phi R_n = 0.75 [0.6 F_y A_{gv} + F_u A_{nt}] = 0.75 [(0.6)(36 \text{ ksi})(3.38 \text{ in}^2) + (58 \text{ ksi})(1.51 \text{ in}^2] = 120.4$ kips

shear rupture, tension yielding, (Equation 7)
$\phi R_n = 0.75 [0.6 F_u A_{nv} + F_u A_{nt}] = 0.75 [(0.6)(58 \text{ ksi})(2.26 \text{ in}^2) + (58 \text{ ksi})(1.51 \text{ in}^2)] = 124.7$ kips

using the upper limit,
$\phi R_n = 120.4$ kips

Unity Check = $(R_u) / (\phi R_n)$ = (80.0 kips) / (120.4 kips) = 0.66 < 1.0, OK

Bolt Shear

factored load, (Equation 8)
$R_u = P_u / N$ = (80.0 k) / (4 bolts) = 20.0 kips / bolt

shear capacity, (Equation 9)
$\phi R_n = 0.75 (0.4 F_u)(A_b)(N_s) = (0.75)(0.4)(120 \text{ ksi})(0.6013 \text{ in}^2)(1.0) = 21.6$ kips / bolt

Unity Check = $(R_u) / (\phi R_n)$ = (20.0 kips / bolt) / (21.6 kips / bolt) = 0.93 < 1.0, OK

Bolt Bearing on Brace

Note: In an actual calculation, this limit state would be skipped because bearing on the gusset has a lower capacity. The gusset is thinner and has a lower material strength.

By inspection, the controlling case is tension because of a higher factored load and shorter edge distances.

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

factored load, (Equation 12)
$R_u = P_u / N = (80\text{ kips}) / (4\text{ bolts}) = 20.0\text{ kips / bolt}$

By inspection, the clear distance is controlled by the edge distance,
$L_c = (1.5\text{ in}) – (7/8\text{ in} + 1/16\text{ in}) /2 = 1.03\text{ in}$

design strength,
$\phi R_n = 0.75\ (1.2)(L_c)(t)(F_u) = (0.75)(1.2)(1.03\text{ in})(0.515\text{ in})(65\text{ ksi}) = 31.0\text{ kips / bolt}$ (Equation 13)

$\phi R_n = 0.75\ (2.4)(d)(t)(F_u) = (0.75)(2.4)(7/8\text{ in})(0.515\text{ in})(65\text{ ksi}) = 52.7\text{ kips / bolt}$ (Equation 14)

using the lessor value,
$\phi R_n = 31.0\text{ kips / bolt}$

Unity Check = $(R_u) / (\phi R_n) = (20.0\text{ kips / bolt}) / (31.0\text{ kips / bolt}) = 0.65 < 1.0$, OK

Bolt Bearing on Gusset

By inspection, the controlling case is tension because of a higher factored load and shorter edge distances.

factored load, (Equation 12)
$R_u = P_u / N = (80.0\text{ k}) / (4\text{ bolts}) = 20.0\text{ kips / bolt}$

design strength,
$\phi R_n = 0.75\ (1.2)(L_c)(t)(F_u) = (0.75)(1.2)(1.03\text{ in})(0.375\text{ in})(58\text{ ksi}) = 20.2\text{ kips / bolt}$ (Equation 13)

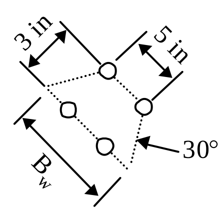
$\phi R_n = 0.75\ (2.4)(d)(t)(F_u) = (0.75)(2.4)(7/8\text{ in})(0.375\text{ in})(58\text{ ksi}) = 34.3\text{ kips / bolt}$ (Equation 14)

using the lessor value,
$\phi R_n = 20.2\text{ kips / bolt}$

Unity Check = $(R_u) / (\phi R_n) = (20.0\text{ kips / bolt}) / (20.2\text{ kips / bolt}) = 0.99 < 1.0$, OK

Gusset Buckling

factored compression load, (Equation 15)
$R_u = P_u = 60.0\text{ kips}$



Whitmore section width based on bolt geometry,
$B_w = (5.00\text{ in}) + (2\text{ sides})(3\text{ in bolt spacing})(\tan 30°) = 8.47\text{ in}$

Whitmore section width based on gusset width,
$B_w = 2\ (6.25\text{ in, w.p. to end of gusset})(\sin 40°) = 8.03\text{ in}$

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

using the lessor value,
B_w = 8.03 in

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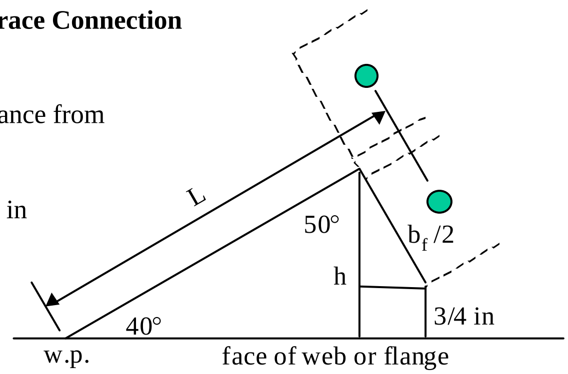
Sample Design - Simple Brace Connection



determine length of Whitmore section using ¾ in clearance from edge of WT to support, refer to diagram at right

h = (3/4 in clearance) + (0.5)(8.005 in)(cos 40°) = 3.81 in

length of whitmore section,
$L = h / \sin 40° +$ bolt edge distance
= (3.81 in) / (sin 40°) + 1.5 in = 7.43 in

$K L / r$ = (1.2)(7.43 in) / [0.375 / sqrt (12)] = 82.4, say 83

from LRFD, Table 4-22, the compressive stress is: ϕF_{cr} = 23.1 ksi

compression capacity, (Reference 16)
$\phi R_n = \phi A_g F_{cr}$ = (8.03 in)(0.375 in)(23.1 ksi) = 69.6 kips

Unity Check = $(R_u) / (\phi R_n)$ = (60.0 kips) / (69.6 kips) = 0.86 < 1.0, OK

Yielding on Gross Section of Gusset

factored tensile load, (Equation 17)
$R_u = P_u$ = 80.0 kips

using the previously determined Whitmore section dimensions, (Equation 18)
$\phi R_n = 0.9 (F_y)(A_g)$ = 0.9 (36 ksi)(0.375 in)(8.03 in) = 97.6 kips

Unity Check = $(R_u) / (\phi R_n)$ = (80.0 kips) / (97.6 kips) = 0.82 < 1.0, OK

Weld Shear

factored brace force,
P_u = 80 kips (tension)

length of weld,
L_w = (6.25 in) + (11.25 in) = 17.5 in (each side of plate)

load eccentricity, (Figure 6)
$e = L_w / 2 - a$ = (17.5 in) /2 – (6.25 in) = 2.5 in

axial component, (Equation 19)
$f_a = P_u (\sin \theta) / 2 L_w + P_u (e) (\sin \theta) 3 / (L_w)^2$
= (80 kips)(sin 40°) / 2 (17.5 in) + (80 kips)(2.5 in)(sin 40°) 3 / (17.5 in)^2 = 2.73 k/in

shear component, (Equation 20)

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

$f_v = P_u (\cos \theta) / 2\,L_w = (80\text{ kips})(\cos 40°) / 2\,(17.5\text{ in}) = 1.75\text{ k/in}$

factored stress, (Equation 21)

$$R_u = \sqrt{(f_a)^2 + (f_v)^2} = \sqrt{(2.73\,\text{k/in})^2 + (1.75\,\text{k/in})^2} = 3.24\,\text{k/in}$$

design strength, (Equation 22)

$\phi R_n = 0.75\,(0.6\,F_{exx})(0.707\,t_w) = 0.75\,(0.6)(70\text{ ksi})(0.707)(0.25\text{ in}) = 5.57\text{ k/in}$

Unity Check = $(R_u) / (\phi R_n) = (3.24\text{ k/in}) / (5.57\text{ k/in}) = 0.58 < 1.0$, OK

Web Check

web width between flanges,

$B_w = d - 2(b_f) = (8\text{ in}) - 2(0.435\text{ in}) = 7.13\text{ in}$

yield line dimension, (Equation 23)

$b = (B_w - c)/2 = (7.13\text{ in} - 0.375\text{ in})/2 = 3.38\text{ in}$

length of plate,

$L = 6.25\text{ in} + 11.25\text{ in} = 17.5\text{ in}$

factored load, (Equation 24)

$R_u = (\text{factored brace force})(\sin \theta) = (80\text{ k})(\sin 40°) = 51.4\text{ kips}$

web resistance, (Equation 25)

$$\phi R_n = 0.9(F_y)(t_w)^2\left[L/b + 4\sqrt{1+(c/2b)}\right]$$

$$= 0.9(50\text{ ksi})(0.285\text{ in})^2\left[\frac{17.5\,\text{in}}{3.38\,\text{in}} + 4\sqrt{1+\frac{0.375\,\text{in}}{2(3.38\,\text{in})}}\right] = 33.9\text{ kips}$$

Unity Check = $(R_u) / (\phi R_n) = (51.4\text{ kips}) / (33.9\text{ kips}) = 1.52$, NG use stiffener plates

Plate Shear

factored load, (Equation 26)

$R_u = (\text{factored brace load})(\cos \theta) = (80\text{ kips})(\cos 40°) = 61.3\text{ kips}$

distance to first bolt,

(0.75 in clear) + (1.5 in end dist)(sin 40°) + (1.5 in edge dist)(cos 40°) = 2.86 in

gusset plate width,

$L = (6.25\text{ in} + 11.25\text{ in}) - (2.86\text{ in}) / (\tan 40°) = 14.1\text{ in}$

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BRACING CONNECTIONS

Sample Design - Simple Brace Connection

plate shear resistance, (Equation 27)
$\phi R_n = (0.9)(0.6)(F_y)(A_g) = (0.9)(0.6)(36\text{ ksi})(0.375\text{ in})(14.1\text{ in}) = 274.1\text{ kips}$

Unity Check = $(R_u) / (\phi R_n)$ = (61.3 kips) / (274.1 kips) = 0.22, OK

Observations

Summary of unity checks,

Fracture on net section of brace, 0.53
Block shear on brace, 0.52
Block shear on gusset, 0.66
Bolt shear, 0.93
Bolt bearing on brace, 0.65
Bolt bearing on gusset, 0.99
Gusset buckling, 0.86
Yielding on gross section of gusset, 0.82
Weld to support, 0.58

The number of bolts is controlled by bolt shear and bolt bearing, fewer bolts will not work.

The brace is controlled by block shear, conceivably a thinner brace flange would be acceptable if such a section is also acceptable for the overall span and load.

The gusset thickness is controlled by bolt bearing, a thinner gusset and larger edge distance would not work due to gusset buckling.

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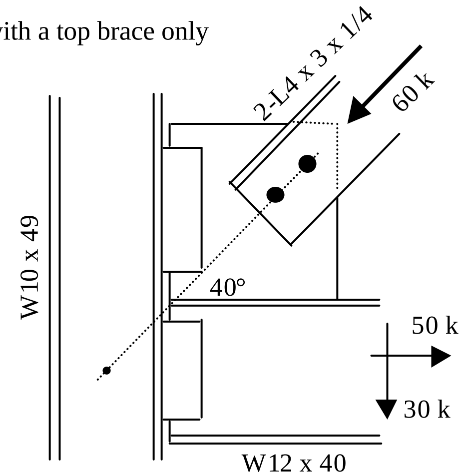
BRACING CONNECTIONS

Sample Design - Combined Brace Connection

Given:

A brace-beam-column connection (standard detail S30) with a top brace only



The following factored load case is to be checked:
Brace Axial Force, P_u = -60 kips (compression)
Beam Shear Force, V_g = 30 kips (downward)
Beam Axial Force, H_g = 50 kips (tension)

Brace: 2 – L 4 x 3 x ¼, LLV, A572, grade 50
angle from horizontal = 40 degrees
b_f = 3.0 in
t_f = 0.25 in
gage = 2 in
F_y = 50 ksi
F_u = 65 ksi

Beam: W12x40, A572, grade 50
d = 11.94 in

Column: W10x49, A572, grade 50
d = 9.98 in

Bolts: 7/8 in diameter, A325
bearing type, threads in shear plane
1.5 in minimum from centerline of bolt to nearest edge of plate
3 inch spacing
A_b = 0.601 in^2 (gross area)
F_u = 120 ksi

Welding: E70XX electrodes, F_{exx} = 70 ksi

Plates: A36
F_y = 36 ksi
F_u = 58 ksi

Try:

Gusset Plate:
2 bolts between brace and gusset plate
1/4 inch fillet weld on each side of gusset plate
L_w = 12 in
5/8 inch gusset plate

Top Shear Plate Connection:
3 bolts, 3 inch spacing

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Sample Design - Combined Brace Connection

¾ in clear from face of column

2 inches from bottom bolt to top of beam

Uniform Force Method

β = (2 inches from T.O. beam to bottom bolt) + (1 each)(3 in bolt spacing) = 5.0 in (Figure 10)

α' = (column clearance) + L_w /2 = (0.75 in) + (12 in) / 2 = 6.75 in (Figure 10)

θ = 40 degrees

e_c = d / 2 = (9.98 in) / 2 = 4.99 in (Figure 10)

e_b = d / 2 = (11.94 in) / 2 = 5.97 in (Figure 10)

$\alpha = e_b \cot\theta - e_c + \beta \cot\theta$ (Equation 28)
$= (5.97\text{ in})(\cot 40°) - (4.99\text{ in}) + (5.0\text{ in})(\cot 40°) = 8.08\text{ in}$

$r = \sqrt{(\alpha + e_c)^2 + (\beta + e_b)^2} = \sqrt{(8.08\text{ in} + 4.99\text{ in})^2 + (5.0\text{ in} + 5.97\text{ in})^2} = 17.06\text{ in}$ (Equation 29)

$H_b = \alpha P_u / r$ = (8.08 in)(-60 kips) / (17.06 in) = -28.42 kips (Equation 30)

$V_b = e_b P_u / r$ = (5.97 in)(-60 kips) / (17.06 in) = -21.00 kips (Equation 31)

$M_b = V_b (\alpha - \alpha')$ = (-21.00 kips)(8.08 in – 6.75 in) = -27.93 in-kips (Equation 32)

$H_c = e_c P_u / r$ = (4.99 in)(-60 kips) / (17.06 in) = -17.55 kips (Equation 33)

$V_c = \beta P_u / r$ = (5.0 in)(-60 kips) / (17.06 in) = -17.58 kips (Equation 34)

Fracture on Net Section of Brace

For this load case, there is no tension in the top brace

Block Shear on Brace

For this load case, there is no tension in the top brace

Block Shear on Gusset

For this load case, there is no tension in the top brace

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BRACING CONNECTIONS

Sample Design - Combined Brace Connection

Brace-Gusset Bolt Shear

factored load, (Equation 8)
R_u = abs (P_u) / N = (60 kips) / (2 bolts) = 30.0 kips / bolt

shear capacity, (Equation 9)
ϕR_n = 0.75 (0.4 F_u)(A_b)(N_s) = (0.75)(0.4)(120 ksi)(0.601 in^2)(2 each) = 43.3 kips / bolt

Unity Check = (R_u) / (ϕR_n) = (30.0 kips / bolt) / (43.3 kips / bolt) = 0.69 < 1.0, OK

Brace-Gusset Bolt Bearing (on brace)

factored load, (Equation 12)
R_u = abs (P_u) / N = (60 kips) / (2 bolts) = 30.0 kips / bolt

By inspection, the load is compression and the clear distance is controlled by bolt spacing,
L_c = (3 in) – (7/8 in + 1/16 in) = 2.06 in

design strength,
ϕR_n = 0.75 (1.2)(L_c)(t)(F_u) = (0.75)(1.2)(2.06 in)(2 x 0.25 in)(65 ksi) = 60.3 kips / bolt (Equation 13)

ϕR_n = 0.75 (2.4)(d)(t)(F_u) = (0.75)(2.4)(0.875 in)(2 x 0.25 in)(65 ksi) = 51.2 kips / bolt (Equation 14)

using the lessor value,
ϕR_n = 51.2 kips / bolt

Unity Check = (R_u) / (ϕR_n) = (30.0 kips / bolt) / (51.2 kips / bolt) = 0.59 < 1.0, OK

Brace-Gusset Bolt Bearing (on gusset)

factored load, (Equation 12)
R_u = abs (P_u) / N = (60 kips) / (2 bolts) = 30.0 kips / bolt

design strength,
ϕR_n = 0.75 (1.2)(L_c)(t)(F_u) = (0.75)(1.2)(2.06 in)(0.625 in)(58 ksi) = 67.2 kips / bolt (Equation 13)

ϕR_n = 0.75 (2.4)(d)(t)(F_u) = (0.75)(2.4)(0.875 in)(0.625 in)(58 ksi) = 57.1 kips / bolt (Equation 14)

using the lessor value,
ϕR_n = 57.1 kips / bolt

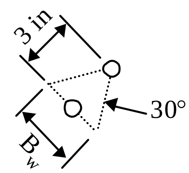
Unity Check = (R_u) / (ϕR_n) = (30.0 kips / bolt) / (57.1 kips / bolt) = 0.53 < 1.0, OK

BRACING CONNECTIONS

Sample Design - Combined Brace Connection

Gusset Buckling

factored load, (Equation 15)
R_u = abs (P_u) = 60 kips



Whitmore section width,
B_w = (2 sides)(3 in bolt spacing)(tan 30°) = 3.46 in

length of Whitmore section is determined from connection dimensions and geometry, L = 5.95 in

for 2 bolt rows, use K = 1.2

K L / r = (1.2)(5.95 in) / [0.625 / sqrt (12)] = 39.5, say 40

from LRFD, Table 4-22, the compressive stress is: ϕF_{cr} = 29.8 ksi

compression capacity, (Reference 16)
$\phi R_n = \phi A_g F_{cr}$ = (3.46 in)(0.625 in)(29.8 ksi) = 63.1 kips

Unity Check = (R_u) / (ϕR_n) = (60.0 kips) / (63.1 kips) = 0.95 < 1.0, OK

Yielding on Gross Area of Gusset

For this load case, there is no tension in the gusset.

Gusset Shear

factored load, (Equation 37)
R_u = abs (H_b) = 28.42 kips

shear capacity, (Equation 38)
ϕR_n = 0.9 (0.6 F_y)(A_g) = 0.9 (0.6)(36 ksi)(12 in)(0.625 in) = 145.8 kips

Unity Check = (R_u) / (ϕR_n) = (28.42 kips) / (145.8 kips) = 0.19 < 1.0, OK

Gusset-Beam Weld Shear

length of weld,
L_w = 12 in

note that gusset to beam forces were computed as part of the unified force method.

for perpendicular forces, substituting V_b for P_u (sin θ), and M_b for P_u (sin θ)(e),
$f_a = V_b / 2 (L_w) + M_b\, 3 / (L_w)^2$ (Equation 19, with substitution)
$= (21.0 \text{ kips}) / 2 (12 \text{ in}) + (27.93 \text{ in-kips})(3) / (12 \text{ in})^2 = 1.457$ k/in

BRACING CONNECTIONS

Sample Design - Combined Brace Connection

for horizontal forces, substituting H_b for P_u (cos θ),
$f_v = H_b / 2 (L_w)$ (Equation 20, with substitution)
$= (28.42 \text{ kips}) / 2 (12 \text{ in}) = 1.184 \text{ k/in}$

$$R_u = \sqrt{(f_v)^2 + (f_a)^2} \quad \text{(Equation 21)}$$
$$= \sqrt{(1.184 \text{ k/in})^2 + (1.457 \text{ k/in})^2} = 1.88 \text{ kips/in}$$

weld capacity, (Equation 22)
$\phi R_n = 0.75 (0.6 F_{exx})(0.707 t_w) = 0.75 (0.6)(70 \text{ ksi})(0.707)(0.25 \text{ in}) = 5.57 \text{ k/in}$

Unity Check = $(R_u) / (\phi R_n) = (1.88 \text{ k/in}) / (5.57 \text{ k/in}) = 0.34 < 1.0$, OK

Shear Plate Design (top plate)

factored loads,
axial, $P_u = \text{abs}(H_c) = 17.55$ kips
shear, $V_u = \text{abs}(V_c) = 17.58$ kips

For brevity, the details of this calculation are not shown. Refer to Practice 000.215.1201, Single Plate Connections.

Shear Plate Design (beam plate)

factored loads,
shear, $V_u = V_g - V_b = (-30 \text{ kips}) – (-21.0 \text{ kips}) = -51.0$ kips (downward) (Equation 35)

axial, $P_u = H_g + H_b = (50 \text{ kips}) + (-28.42 \text{ kips}) = +21.58$ kips (tension) (Equation 36)

For brevity, the details of this calculation are not shown. Refer to Practice 000.215.1201, Single Plate Connections.

Observations

Summary of unity checks,

Brace-Gusset Bolt Shear, 0.69
Brace-Gusset Bolt Bearing (on brace), 0.59
Brace-Gusset Bolt Bearing (on gusset), 0.53
Gusset Buckling, 0.95
Gusset Shear, 0.19
Gusset-Beam Weld Shear, 0.34

Because there are only two bolts, fewer bolts will not work. A thinner angle thickness is possible, however the angle must be checked for overall span and load.

FLUOR®

BRACING CONNECTIONS

Sample Design - Combined Brace Connection

The gusset thickness is controlled by buckling. The weld size is controlled by a minimum size of ¼ inch.